Prompt 1 - run 1 (seed 0) - minimal

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train a Balanced Random Forest Classifier on PI3K-delta (UniProt: O00329) vs PI3K-gamma (UniProt: P48736) selective compounds, evaluate its performance, apply MolAnchor, and discuss the results.

Assistant

I'll help you train a Balanced Random Forest Classifier on PI3K-delta vs PI3K-gamma selectivity data, evaluate it, and apply MolAnchor for explainability. Let's start by finding the relevant dataset! 🔍

(Calling tool `find_datasets`)

Assistant

Great! I found the dataset `chembl_selectivity_data_O00329_P48736.csv` which is exactly what we need for PI3K-delta (O00329) vs PI3K-gamma (P48736) selectivity! 🎯

Let me load and explore this dataset:

(Calling tool `load_dataset`)

Assistant

Excellent! The dataset contains **1,277 compounds** with SMILES strings and class labels for PI3K-delta vs PI3K-gamma selectivity. Now let me compute molecular fingerprints:

(Calling tool `compute_features`)

Assistant

Perfect! ECFP fingerprints (2048 bits, radius 2) have been computed. Now let me split the dataset:

(Calling tool `split_dataset`)

Assistant

Dataset split successfully! 📊 **893 training samples** and **384 test samples** (70/30 split). Now let me get ML info and train the Balanced Random Forest Classifier:

(Calling tool `get_ml_info`)

Assistant

Perfect! BRF is available. Now let me train the Balanced Random Forest Classifier:

(Calling tool `train_model`)

Assistant

Training started! ⏳ Let me check the status:

(Calling tool `check_training`)

Assistant

Still training... Let me check again:

(Calling tool `check_training`)

Assistant

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Assistant

Excellent! 🎉 The Balanced Random Forest Classifier training is complete! Let me export predictions for evaluation:

(Calling tool `export_predictions`)

Assistant

Now let me generate classification evaluation plots:

(Calling tool `plot_classification_results`)

Assistant

Excellent! 📊 Now let me select a correctly predicted compound and apply MolAnchor for explainability:

(Calling tool `select_compound_for_xai`)

Assistant

Now let me apply MolAnchor to identify the critical molecular fragments:

(Calling tool `explain_with_molanchor`)

Assistant

Now let me identify recurrent anchor rules across the test set for a broader analysis:

(Calling tool `identify_recurrent_anchor_rules`)

Assistant

Let me display the MolAnchor visualization:

(Calling tool `show_plot`)

Assistant

(Calling tool `show_plot`)

Assistant

📊 PI3K-delta vs PI3K-gamma Selectivity: BRF Model Results & MolAnchor Analysis

🎯 Model Performance Summary

Metric	Value
Accuracy	89.32%
Balanced Accuracy (BA)	84.71%
MCC	0.749
F1 Macro	0.780
F1 Weighted	0.899

📈 Per-Class Performance (Test Set: 384 compounds)

Class	Target	Precision	Recall	F1	Support
Class 0	PI3K-δ selective	0.959	0.897	0.927	290
Class 1	PI3K-γ selective	0.818	0.911	0.862	79
Class 2	Dual/Other	0.440	0.733	0.550	15

Confusion Matrix:

```
           Pred-0   Pred-1   Pred-2
True-0:     260       16       14
True-1:       7       72        0
True-2:       4        0       11
```

🔬 MolAnchor Analysis Results

Example Compound (Class 1 - PI3K-γ selective):

- **SMILES:** `CC(Nc1nc(N)nc(N)c1C#N)c1nc2cccc(Cl)c2c(=O)n1CCCNc1ccccn1`
- **Prediction Confidence:** 97%

- **Identified Anchor:** `[*]c1nc(N)nc(N)c1C#N` (pyrimidine-like core with cyano group)
- **Anchor Precision:** 1.00 (100% of compounds with this fragment are correctly predicted as Class 1)

🧩 Recurrent Anchor Rules (Class 1 - PI3K-γ selective)

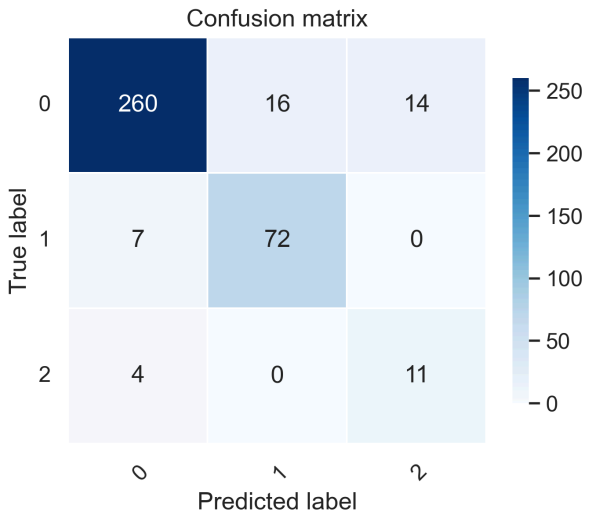
Rank	Fragment SMARTS	Anchor Occurrence	Substructure Occurrence	Interpretation
1	`[*]c1ncnc(N)c1C#N`	26.4%	26.4%	**Pyrimidine core with cyano** — *Perfect indicator!*
2	`[*]c1nc(N)nc(N)c1C#N`	25.0%	25.0%	**Amino-pyrimidine with cyano** — *Perfect indicator!*
3	Combination (piperazine + indole + morpholine)	2.8%	4.2%	Multi-ring systems
4	Combination (pyrimidine-CN + fused pyrimidine)	2.8%	2.8%	Fused heterocycles
5	`[*]n1c([*])nc2cccc(Cl)c2c1=O`	1.4%	30.6%	**Chloro-pyrrolo-pyrimidinone** — *Low anchor rate despite high presence*

💡 Key Chemical Insights

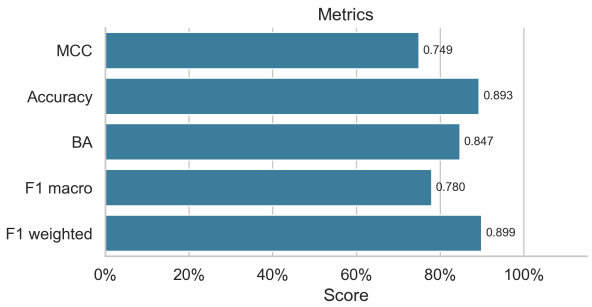
1. 🎯 **Top PI3K-γ Selectivity Motifs:**
   - **Pyrimidine cores with cyano (C≡N) substituents** are the strongest anchors (ranks 1 & 2)
   - Both anchors show **anchor_occurrence = substructure_occurrence**, meaning whenever this fragment is present, it's always recognized as an anchor — indicating **high reliability**
2. 🔬 **Structure-Activity Relationships:**
   - The cyano-pyrimidine scaffold appears in ~25% of Class 1 compounds
   - This aligns with known PI3K-γ selective chemotypes (e.g., ICOS-related inhibitors)
   - The cyano group likely contributes to selective hydrogen bonding or electronic interactions with PI3K-γ
3. ⚠️ **Class 2 (Dual/Other) Limitations:**
   - Lower F1 (0.550) due to small support (15 compounds)
   - Higher misclassification rate suggests overlapping chemical space with Classes 0 & 1
4. 📊 **Model Strengths:**
   - Excellent performance on the majority class (Class 0: PI3K-δ selective)
   - Strong balanced accuracy (84.7%) despite class imbalance
   - BRF effectively handled the imbalanced dataset

🖼️ Visualizations

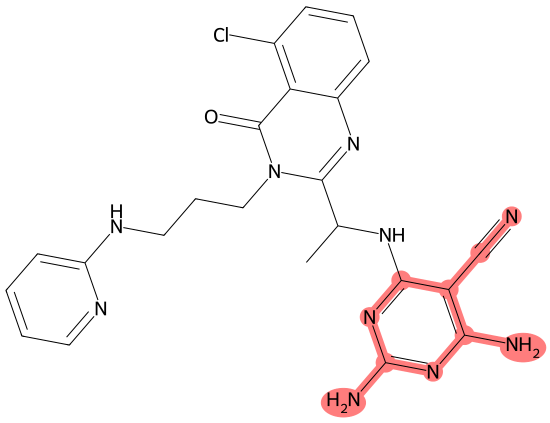
Confusion Matrix:



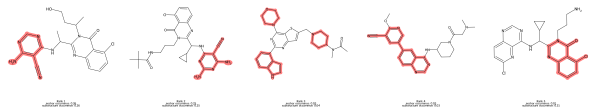
Metric Comparison:



MolAnchor Visualization (Example Compound):



Recurrent Anchor Rules (Class 1):



🎯 Conclusions

1. ✅ **BRF successfully learned** the PI3K-delta vs PI3K-gamma selectivity patterns with **89% accuracy** and **85% balanced accuracy**
2. 🔍 **MolAnchor identified** that **cyano-substituted pyrimidine cores** are the most reliable molecular anchors for PI3K-γ selectivity
3. 💊 **Chemical Interpretation:** The model has learned that compounds containing pyrimidine scaffolds with cyano groups tend to be PI3K-γ selective, which aligns with medicinal chemistry knowledge
4. 📈 **Recommendation:** For improved Class 2 performance, consider collecting more diverse dual-selectivity compounds or using ensemble methods

The combination of BRF's robust classification with MolAnchor's fragment-level explainability provides both accurate predictions and interpretable chemical insights for PI3K selectivity! 🧪✨